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ORIGINAL

The effect of acute sleep deprivation on cortisol level: a systematic review and meta-analysis

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Abstract. Acute sleep deprivation has aroused widespread concern and the relationship between acute sleep deprivation and cortisol levels is inconsistent. This study aimed to explore additional evidence and details. The PubMed, Web of Science, EMBASE, CLINAHL and Cochrane databases were searched for eligible studies published up to June 7, 2023. All analyses were performed using Review Manager 5.4 and Stata/SE 14.0. A total of 24 studies contributed to this meta-analysis. There was no significant difference in cortisol levels between participants with acute sleep deprivation and normal sleep in 21 crossover-designed studies (SMD = 0.18; 95% CI: –0.11, 0.45; $p = 0.208$) or 3 RCTs (SMD = 0.26; 95% CI: –0.22, 0.73; $p = 0.286$). Subgroup analysis revealed that the pooled effects were significant for studies using serum as the sample (SMD = 0.46; 95%CI: 0.11, 0.81; $p = 0.011$). Studies reporting cortisol levels in the morning, in the afternoon and in the evening did not show significant difference ($p > 0.05$). The pooled effects were statistically significant for studies with multiple measurements (SMD = 0.28; 95%CI: 0.03, 0.53; $p = 0.027$) but not for studies with single cortisol assessments ($p = 0.777$). When the serum was used as the test sample, the cortisol levels of individuals after acute sleep deprivation were higher than those with normal sleep.

Key words: Acute sleep deprivation, Cortisol, Meta-analysis

ADEQUATE SLEEP is essential for maintaining good health and well-being and has received considerable attention owing to the swiftness of societal development and increased recognition. The National Sleep Foundation recommends an average of 9–11 hours of sleep for school-aged children, 8–10 hours for teenagers and 7–9 hours for young adults and adults [1, 2]. However, sleep deprivation seems to be prevalent [3], with acute sleep deprivation defined as “the condition associated with a complete absence of sleep during the previous night or nights” [4]. A study showed that only 34.41% of high school students in China had ≥8 hours of sleep [5]. The other reported that shift workers frequently had shortened sleep duration [6], and more than half of the staff nurses suffered from sleep deprivation, which caused more patient care errors [7]. Acute sleep deprivation is becoming a prevalent public health concern because of its adverse effects on human health and quality of life, such as an increased the risk of cardiovascular disease [8], reduced respiratory motor performance [9], inducing cognitive deficits [10] and linking with immune deregulation [11]. People experience acute sleep deprivation owing to many factors, including aging, stress and sleep disorders [12-14].

In humans, cortisol (11b,17a,21-trihydroxypregn-4-ene-3,20-dione) is a key glucocorticoid in the endocrine and stress response systems and is secreted *via* the hypothalamic-pituitary-adrenal (HPA) axis [15]. The central control of the HPA axis is very complex and vital for the neuroendocrine system, which regulates the rhythm of daily hormone secretion and influences many physiological functions to respond appropriately to the changing environments [16]. Cortisol is the end product of the HPA axis and displays the same circadian rhythm [17]. Its levels experience a rapid increase in the first 30–40 min of awakening, reaching a peak known as the cortisol awakening response; subsequently, the level

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gradually declines throughout the day until reaching the nadir at midnight [18-20], during which time the HPA axis activity is also at its minimum [17]. Therefore, this level has a wide range. Some studies have shown that abnormally high cortisol levels are associated with adverse outcomes, such as obesity, type 2 diabetes, depression, and sleep apnea [17, 21-23].

Notably, some studies have proposed that acute sleep deprivation could cause cortisol levels to increase, whereas the others have reached the opposite conclusion. Based on the harmful effects of poor sleep quality and increased cortisol levels on human health, coupled with the inconsistent findings, this study aimed to investigate the association between acute sleep deprivation and cortisol levels using a systematic review and meta-analysis.

Materials and Methods

Literature search

Until June 7, 2023, a systematic search was conducted on five electronic databases, PubMed, Web of Science, EMBASE, Cumulative Index to Nursing and Allied Health Literature (CINAHL) and Cochrane, for articles without any filters. The search strategy for PubMed was as follows: (((((Deprivation, Sleep) OR (REM Sleep Deprivation)) OR (Sleep Insufficiency)) OR (Inadequate Sleep)) OR (Sleep Fragmentation)) AND ((((Hydrocortisone) OR (Cortisol)) OR (Cortef)) OR (Cortril)). The details of search strategy for the other databases were showed in Supplementary Table 1. Before conducting the review, we had a written protocol that included all the following: a review question, a search strategy, inclusion/exclusion criteria, and a risk of bias assessment.

Inclusion and exclusion criteria

Original and peer-reviewed articles, including those with a randomized crossover design, randomized clinical trials (RCTs), and clinical trials were eligible for inclusion. The Population, Intervention, Comparison and Outcome (PICO) characteristics for eligibility were as follows: (1) healthy adults (*i.e.*, 18 + years old) with normal sleep and without glucocorticosteroids use; (2) acute sleep deprivation intervention was performed; (3) controls with normal sleep duration; (4) measurements of cortisol concentration (salivary, serum and plasma samples).

The exclusion criteria were as follows: (1) reviews, conference proceedings and meta-analysis articles; (2) animal or *in-vitro* experiments; (3) inability to obtain full text and raw data not available after contacting the author; and (4) non-English articles.

Data extraction

A data extraction form was pre-designed for the eligible articles, which included the first author’s name, year of publication, country, study design, number of males and females, percentage of males, mean age or age range, type of sample, time of collection and method of assessment. The full texts were independently reviewed by two reviewers, data was extracted when they agreed on selection of eligible studies and achieved consensus on which studies to include, a third reviewer was consulted if there was no agreement.

Quality assessment

All eligible studies were assessed by two independent reviewers using the Cochrane risk-of-bias tool [24]. It assessed the quality of RCTs in six domains (selection bias, performance bias, detection bias, attrition bias, reporting bias and other bias) by categorizing each area as “low risk,” “unclear” or “high risk.”

Data synthesis and statistical analysis

In some studies, the cortisol levels were reported as standard error of the mean (SEM) and were converted to standard deviation (SD), (SD = SEM × $\sqrt{N}$; N = number of individuals).

The standardized mean difference (SMD) and 95% confidence interval (CI) were calculated to assess the data; the significance of the pooled SMD was assessed using the Z-test. Heterogeneity was determined by chi-squared and I^2 statistics, ranging from 0 to 100%. $p < 0.1$ was considered significant and $I^2 \leq 60\%$ credible. The first choice for calculating pooled estimates was random-effects model. Subsequently, a sensitivity analysis was performed to detect the impact of individual studies on the overall effects by omitting every single article. For subgroup analysis, sample type (plasma, serum, saliva), collection time (all day, morning: 7:00–12:30, afternoon: 12:30–20:00, night: 20:00–7:00) and number of cortisol measurements (once or the average of multiple time-points) were performed to explore potential heterogeneity. This analysis indicated whether there were significant associations of study publication year, the country where the participants was located, mean age, method of assessment and proportion of men with the pooled SMD. Funnel plots and Egger’s test were conducted to determine publication bias. Egger’s test provides a linear regression between the precision of the studies and the standardized effect. A quality assessment was conducted using Review Manager 5.4 and the remaining statistical analysis was performed using Stata/SE 14.0, $p < 0.05$ was considered significant.

Results

Literature search

Based on the search strategy, 2,490 papers were initially identified from 5 databases. After removing duplicates, then screening the titles and abstracts, study type and data availability, 34 papers were included for full-text screening. Among these remaining papers, 10 papers were excluded because of their unqualified outcomes and groups, unknown sample collection time, or irretrievable data. Thus, 24 papers were finally included for meta-analysis, 3 of which were RCTs and the rest were crossover designs (Fig. 1).

Study characteristics and quality assessment

The characteristics of each study are summarized in Table 1. All the selected papers were published between 1980 and 2022. Among the 24 studies, 3 were RCTs and 21 were crossover-designed papers. The mean age of participants ranged from 19 to 46 and the sample size ranged from 7 to 218. The majority of these articles were from Europe [25-35] (n = 11), with the other 14 studies were conducted in Asia [36-40] (n = 5), America [41-47] (n = 7), and Oceania [48] (n = 1). Of these, 18 papers were conducted exclusively on males and there were only 6 with more than half of the participants being women.

Regarding cortisol collection time, 11 studies examined cortisol levels at multiple time points throughout the day [25, 28, 30, 31, 33, 35, 36, 44, 45, 47, 48], the rest studies collected cortisol once or twice in the morning, afternoon or at night. Regarding the sample type of cortisol, 13 studies used plasma [27, 28, 30-35, 37, 45-48], 3 used saliva [36, 41, 43] and 8 used serum [25, 26, 29, 38-40, 42, 44].

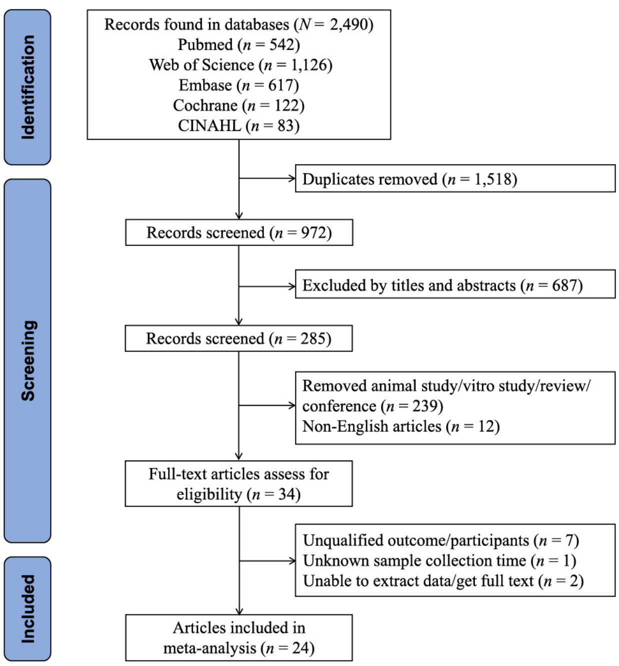


Fig. 1 Flow diagram of the literature search process

An assessment of the quality of included papers is illustrated in Fig. 2. It indicates that most of these papers were categorized as having relatively low risk. However, the biases of some papers remained unclear, particularly in terms of selection, performance and detection reasons. Overall, the risk of bias was low and some of the included study results were largely consistent, while the others were not. The results were precise with small margins of error.

Overall meta-analysis

For crossover-designed papers, significant heterogeneity was found among these studies (I^2 = 76.6%, p < 0.001); consequently, a random-effects model was selected. As shown in Fig. 3, there were no significant differences in cortisol levels between normal or acute sleep-deprived subjects (SMD = 0.18; 95% CI: –0.11, 0.45; p = 0.208). For RCTs, as shown in Fig. 4, there was no significant heterogeneity (I^2 = 0.00%, p = 0.679) or no significant differences in cortisol levels of normal or acute sleep-deprived subjects (SMD = 0.26; 95% CI: –0.22, 0.73; p = 0.286).

Sensitivity analysis

Sensitivity analysis of RCTs and crossover-designed studies are shown in Supplementary Figs. 1 and 2, respectively; they might show that omitting individual articles had little effect on the results of the combined effect size.

Subgroup analysis

Due to significant heterogeneity, subgroup analyses were performed only for crossover-designed studies. The results are shown in Fig. 5. In Fig. 5a, the pooled effects were significant for studies using serum as the sample (SMD = 0.46; 95%CI: 0.11, 0.81; p = 0.011), but not for plasma and saliva (Plasma: SMD = –0.04; 95%CI: –0.52, 0.45; p = 0.882. Saliva: SMD = –0.20; 95%CI: –1.70, 1.31; p = 0.796). Cortisol collection time (Fig. 5b) exhibited no significant subgroup difference. Studies reporting cortisol levels in the morning (SMD = 0.09; 95%CI: –0.20,0.39; p = 0.518), in the afternoon (SMD = 0.26; 95%CI: –0.02, 0.55; p = 0.064) and in the evening (SMD = 0.20; 95%CI: –0.14, 0.53; p = 0.255) did not show significant difference. Regarding the number of measurements (Fig. 5c), the pooled effects were statistically significant for studies with multiple measurements (SMD = 0.28; 95%CI: 0.03, 0.53; p = 0.027) but not for studies

Table 1 Characteristics of included studies

Author (year)	Country	Study Design	Male/ Female	Male Percentage (%)	Age (mean ± SD)	Type of sample (collection time)	Method of assessment	Sleep measure	Explanation of outcomes	Source of funding
Torbjorn Akerstedt (1980)	Sweden	crossover design	12/0	100	Range: 19–30	plasma (morning)	radioimmunoassay	—	Cortisol varied significantly over time; The levels of cortisol was significantly lower at the end of the sleep deprivation period.	Nova Southeastern University President's Faculty Research and Development Grant (#33582)
V. Cortes -Gallegos (1982)	Mexico	crossover design	13/0	100	Range: 21–26	plasma (morning)	—	—	Regarding cortisol, there were no significant differences in the individual values found in this population before and after the 24 hr restless period.	Deutsche Forschungsgemeinschaft (SFB654 "Plasticity and Sleep"), Germany, and the Swedish Research Council, Sweden
Janet Mullington (1996)	Germany	RCT	32/0	100	24.7 ± 0.5	plasma (multiple time points 20:00–8:00^{+1})	radioimmunoassay	Observation	Cortisol levels were significantly higher in the sleep-deprivation group.	Swedish Brain Research Foundation (JC, CB), AFA Försäkring (CB), the Novo Nordisk Foundation (CB, JZ), the Swedish Society ofMedicine (JC), Magnus Bergvall's Foundation (JC), Thu ring's Foundation (JC), Tore Nilsson's Foundation (JC) and the Swedish Research Council (HS, JZ).
Rachel Leproult (1997)	America	RCT	26/0	100	Range: 20–32	plasma (multiple timepoints 18:00–2:00^{+2})	radioimmunoassay	Polygraphic sleep recording	After partial and total sleep deprivation, plasma cortisol levels over the 1,800–2,300-hour period were higher on day 2 (sleep deprivation) than on day 1 (control day).	—
Victor Hng-Hang Goh (2001)	Sinapore	RCT	14/0	100	Range: 20–30	saliva (multiple timepoints 8:00–18:00^{+1})	scintillation proximity assay	—	Significant increases in cortisol was noted, especially at 1:30 p.m., on the day after nighttime sleep deprivation.	UL1 TR000135 from the National Center for Advancing Translational Sciences (NCATS)
Florian Chapotot (2001)	France	crossover design	10/0	100	Range: 21–27	plasma (multiple timepoints 23:00–18:00^{+1})	radioimmunoassay	Polysomnography	Mean cortisol plasma levels and secretory rate between 23.00 and 18.00 h were also significantly increased.	the Max Planck Society for postdoctoral fellowship
P. Schüssler (2006)	Germany	crossover design	5/3	62.5	26.3 ± 4.4	plasma (multiple timepoints 23:00–7:00^{+1})	radioimmunoassay	Polysomnography	ACTH and cortisol were also elevated, which was most pronounced during the second half of the night.	the National Aeronautics and Space Administration (NASA) [grant numbers NNX14AN49G and 80NSSC20K0243 (to NG)] and National Institutes of Health [grant number NIH R01DK117488 (to NG)]

Table 1 Cont.

Author (year)	Country	Study Design	Male/ Female	Male Percentage (%)	Age (mean ± SD)	Type of sample (collection time)	Method of assessment	Sleep measure	Explanation of outcomes	Source of funding
J. S. Costa Ricardo (2008)	England	crossover design	11/0	100	20 ± 3	plasma (morning)	enzyme linked immunosorbent assay	Accelerometer	Plasma cortisol concentration increased post-TT (62%) compared with post-SS.	Norwegian Naval Special Operations Command (NORNAVSOC)
Christian Benedict (2011)	Germany	crossover design	14/0	100	22.6 ± 2.99	serum (multiple timepoints 18:00–18:00^{+1})	chemiluminescence immunoassay	Polysomnography	Nocturnal wakefulness increased nocturnal and daytime circulating concentrations of cortisol	the French Délégation Générale pourl'Armement (Contract No. 08co704)
M. Chennaoui (2011)	France	crossover design	12/0	100	29.1 ± 3.3	plasma (multiple timepoints 8:00–23:00)	radioimmunoassay	Wrist actigraphy	There were no significant changes in cortisol with an acute sleep deprivation.	—
Pavel Kavčič (2011)	Slovenia	crossover design	7/0	100	Range: 25–35	plasma (multiple timepoints 23:00–19:00^{+1})	radioimmunoassay	Polysomnography	During the 56-hour sampling period, all participants showed an expected diurnal variation in plasma cortisol. Significant diurnal variation was found also under baseline and sleep deprived conditions.	Associação Fundo de Incentivo à Pesquisa (AFIP), Fundo de Amparo à Pesquisa do Estado de São Paulo (FAPESP–protocols 2010/07005-0 and CEPID #98/14303-3), and CNPq (142060/2012-7)
Masayuki Konishi (2013)	Japan	crossover design	10/0	100	22.6 ± 2.2	plasma (morning, afternoon)	radioimmunoassay	Actiwatch	Plasma cortisol concentrations was not significantly different between trials at any sampling time.	Grant-in-Aid for the Global COE Program 'Sport Science for the Promotion of Active Life' (2010–2011) and a Waseda University Grant for Special Research Projects (Project number: 2011B-283)
Xin-Yang Sun (2013)	China	crossover design	218/0	100	20.77 ± 2.08	serum (morning)	radioimmunoassay	Sleep diary	Cortisol levels after sleep deprivation was significantly higher than that before sleep deprivation.	—
Jonathan Cedernaes (2015)	Sweden	crossover design	15/0	100	22.3 ± 1.9	serum (morning)	enzyme immunoassay	Polysomnography	Following acute sleep deprivation, fasting serum cortisol concentrations were decreased at 0730h, compared with after sleep.	—
Hong-tao Song (2015)	China	crossover design	148/0	100	20.52 ± 1.80	serum (morning)	radioimmunoassay	Sleep diary	Sleep deprivation could significantly increase serum cortisol level.	—

Table 1 Cont.

Author (year)	Country	Study Design	Male/ Female	Male Percentage (%)	Age (mean ± SD)	Type of sample (collection time)	Method of assessment	Sleep measure	Explanation of outcomes	Source of funding
Kenneth P. Wright Jr. (2015)	America	crossover design	14/3	82.35	31.7 ± 6.1	plasma (multiple timepoints 24:00–24:00^{+1})	chemiluminescence immunoassay	Wrist actigraphy	Acute total sleep deprivation significantly increased cortisol levels.	NASA Cooperative Agreement NCC9-58 with the National Space Biomedical Research Institute and NASA, by NIH RO1 HL081761 and R21 DK092624
Yavuz Selvi (2015)	Turkey	crossover design	16/16	50	male: 24 female: 27	serum (morning)	chemiluminescence immunoassay	Observation	There hasn't been any significant statistical change determined on cortisol levels after sleep deprivation.	—
Havard Hamarsland (2018)	Norway	crossover design	15/0	100	23 ± 4	serum (9:00)	—	—	Cortisol increased during the sleep deprivation (0 h) and stayed elevated at 72 h and 1 wk after the sleep deprivation.	Deutsche Forschungsgemeinschaft (Ste 486/5-3)
Murilo Dáttilo (2019)	Brazil	crossover design	10/0	100	24.5 ± 2.9	serum (multiple timepoints 19:00–19:00^{+1})	chemiluminescence immunoassay	Polysomnography	Cortisol was higher in DEPRIVATION.	—
Tom Cullen (2019)	England	crossover design	10/0	100	27 ± 6	plasma (morning)	enzyme linked immunosorbent assay	Actigraphy	Resting concentrations of cortisol were not impacted by sleep deprivation.	Fideicomiso IMSS-FORD FOUNDATION
Peter Y. Liu (2020)	America	crossover design	17/0	100	24.1 ± 2.9	serum (6:00–9:00, 15:00–18:00)	chemiluminescence immunoassay	—	Sleep deprivation increased cortisol in the afternoon in older men.	U.S. National In stitutes of Health grant DK-41814, U.S. Air Force Office of Scientific Research grant AFOSR-94-1-0203, Mind-Body Network of the MacArthur Foundation (Chica go, IL), and the Belgian Fonds de la Re cherche Medicale.
Erika M. Yamazaki (2021)	America	crossover design	18/14	56.25	35.1 ± 7.1	saliva (8:00, 17:00)	enzyme linked immunosorbent assay	Actigraphy	Cortisol changed significantly across the study, increasing during acute total sleep deprivation.	—
Séverine Lamon (2021)	Australia	crossover design	7/6	53.85	Range: 18–35	plasma (multiple timepoints 7:00–16:00)	enzyme linked immunosorbent assay	Actigraphy	Sleep deprivation increased plasma cortisol by 21%.	Yüzüncü Yıl University Scientific Research Project (Project No: 2010-TF-U143)
Thompson, Kayla I (2022)	America	crossover design	14/9	60.87	20.78 ± 2.87	saliva (7:00–9:00)	enzyme immunoassay	Actiwatch wrist monitor	Sleep deprivation results in decreased cortisol levels in the morning	Republic of Slovenia Research Agency, grant No. P3-0338

Fig. 2 Graph (a) and summary (b) of the risk of bias

with single cortisol assessments (SMD = 0.06; 95%CI: –0.38, 0.51; $p = 0.777$).

Publication bias

The funnel plots of RCTs and crossover-designed studies are presented in Figs. 6 and 7. However, the funnel plots were examined by visual inspection to determine the asymmetry. Therefore, Egger's test was performed to statistically assess the publication bias. As shown in Supplementary Figs. 3 and 4, they did not identify any publication bias (RCTs: $p = 0.411$; crossover-designed studies: $p = 0.901$).

Discussion

Sleep problems are closely related to human health and are garnering increasing attention. A total of 24 articles collected from 5 databases were included in this systematic review and meta-analysis. We found that there was no significant difference in total cortisol levels between participants after acute sleep deprivation and those with normal sleep. However, in subgroup analysis, higher serum cortisol levels were observed in individuals with acute sleep deprivation than in those with normal sleep.

Cortisol release is regulated by the HPA axis, which is the central stress response system in humans. After production, cortisol enters the bloodstream and distributed in various tissues, such as saliva, hair, *etc*. Notably, there is a negative and non-linear feedback between cortisol and ACTH release [49].

Multiple methods exist for determining cortisol levels

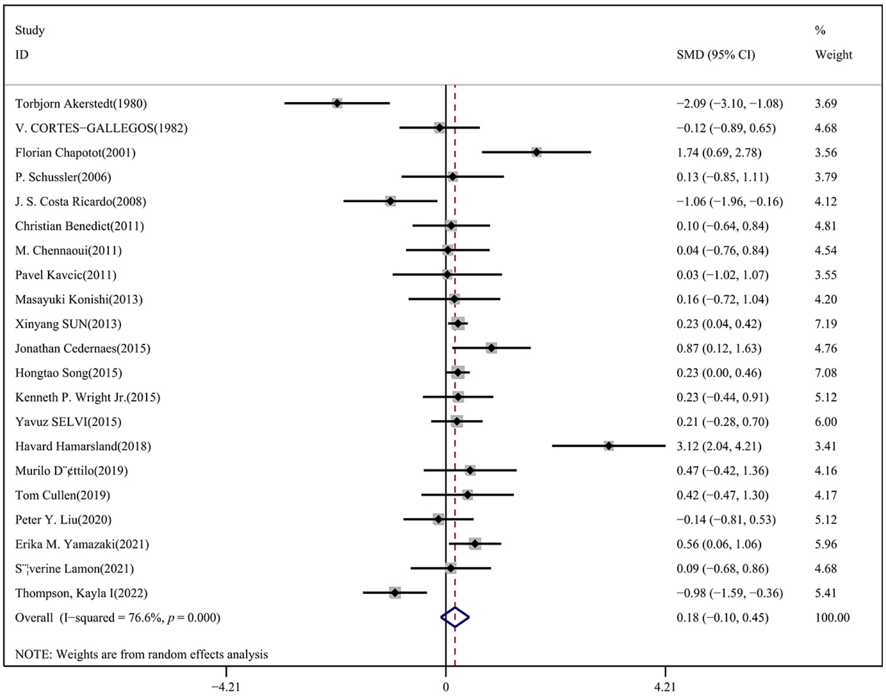


Fig. 3 Forest plot of cortisol level in normal sleep and sleep-deprivation people in crossover-designed studies

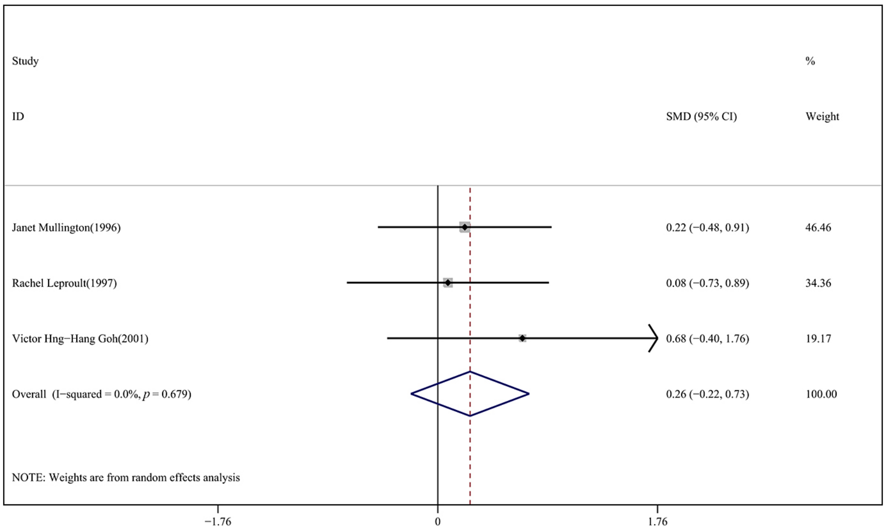


Fig. 4 Forest plot of cortisol level in normal sleep and sleep-deprivation people in RCTs

in biological matrices, such as hair, blood, urine, sweat, and saliva [50]. Only unbound cortisol is biologically active, so it is superior to total cortisol quantification for detecting free cortisol levels. The samples of included studies were plasma, serum and saliva. Salivary cortisol is highly correlated with blood cortisol [51], but the reference value of them was not the same in healthy people, blood cortisol is 25 μg/mL in the morning and 2 μg/mL in the evening [52], salivary cortisol is 1–12 ng/mL in the morning and 0.1–3 ng/mL in the evening [53]. From this reference value, we can find that salivary cortisol level is lower than blood cortisol and research has shown that cortisol levels in blood are approximately 100-fold higher than those in saliva [54]. Salivary samples are easily available and non-invasive, and the level is less susceptible to factors such as pregnancy status and diuretic use. However, they contain many interfering substances, such as proteins, mucus, food debris and so on [50].

To our knowledge, glucocorticoids secreted by the adrenal glands follow a circadian rhythm and are produced at higher levels during wakefulness (daytime in

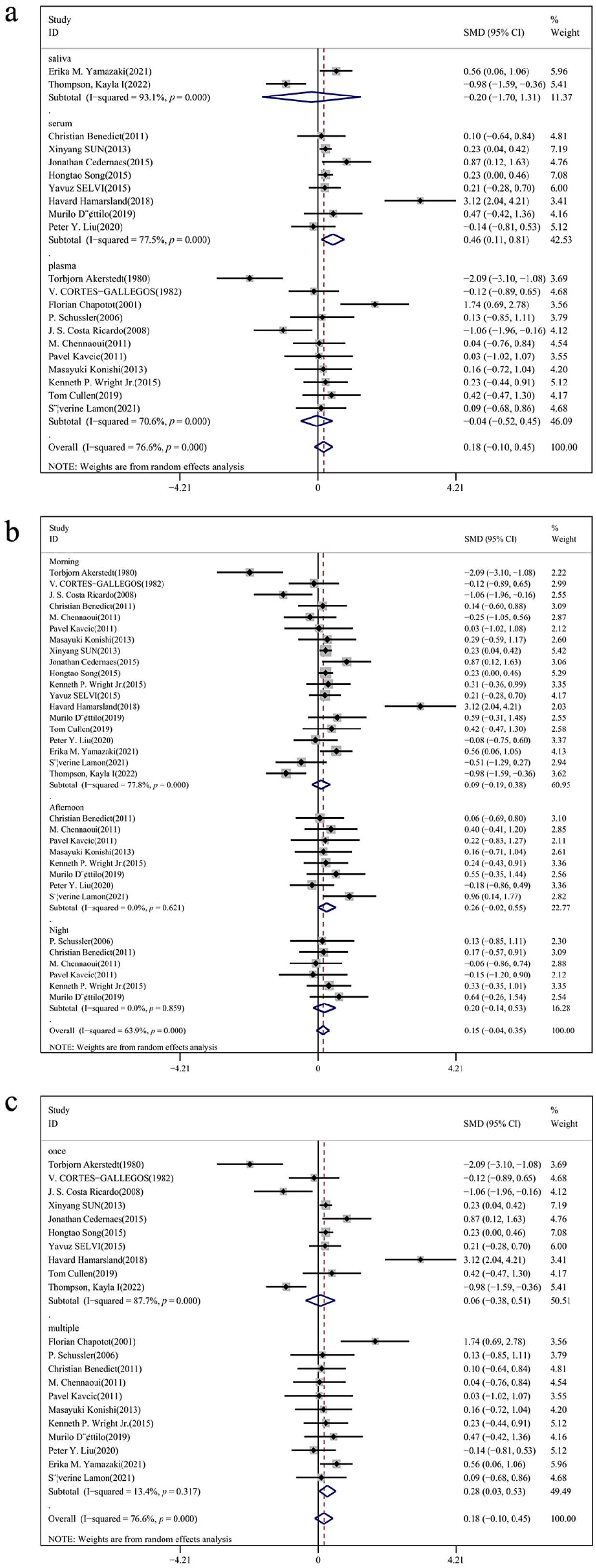


Fig. 5 Subgroup analysis of cortisol level based on sample type (a), collection time (b) and number of measurements (c) in crossover-designed studies

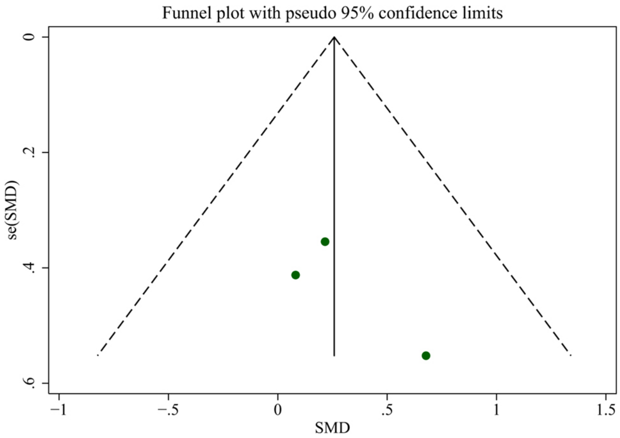


Fig. 6 Funnel plot of RCTs

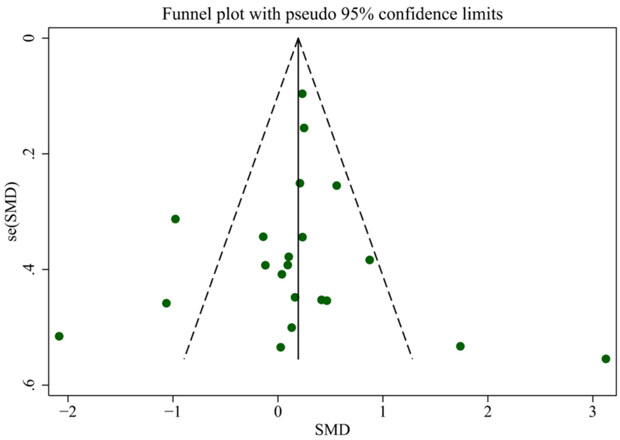


Fig. 7 Funnel plot of crossover-designed studies

humans and nighttime in rodents). In this study, we did not observe that cortisol levels of acute sleep deprivation people were significantly higher or lower than those of normal sleep ones in the morning, afternoon or evening. This conclusion is inconsistent with previous studies [55, 56], which revealed that greater sleep loss characterized by 4 h of sleep for one to six consecutive nights was related to increased cortisol level in the afternoon. Prolonged sleep deprivation leads to increased corticosteroid and ACTH levels, promoting cortisol release [57]. Another animal experiment demonstrated lower cortisol levels in the control group than in the sleep deprivation group in 56 male Wistar rats [58]. In addition to the circadian rhythm, there is an oscillatory pattern of glucocorticoid secretion with an hourly ultradian rhythm. The rhythm of glucocorticoid secretion and its rapid increase in response to stimulation are important for maintaining homeostasis and adapting to stress. The amplitude of the secretion peak varies with the time of the day, whereas the elevation of the secretion value during awake activity is determined by an increase in the amplitude of the pulse.

This study had several advantages. Many studies have focused on the association between cortisol and depression [23], cortisol and stress [14]. However, few studies paid attention to sleep diseases and acute sleep deprivation. In this systematic review and meta-analysis, we assessed the impact of acute sleep deprivation not only on the overall cortisol level, but also on the sample type (plasma, serum and saliva), collection time (all day, morning, afternoon and night) and method of cortisol assessment (*i.e.*, enzyme linked immunosorbent assay *etc.*) of cortisol. These findings provide a comprehensive understanding of how sleep deprivation affects cortisol level in human body.

However, this review still had some limitations. First, non-English papers were not included because, after searching, we discovered that papers in English accounted for a large proportion and translators corresponding to the languages could not be found in a short time, so we cannot guarantee the quality of the translated papers. Second, the methods used to measure cortisol levels differed. These different detection methods have high accuracy and stability; however, in actual operation, there are many influencing factors, complicated steps, and high requirements for operators, which may affect the outcomes. Third, most of the participants in these included studies were men, and more samples are needed to validate the results if the conclusion applies to women. Fourth, the included studies were not all RCTs, so the conclusion needs to be carefully interpreted.

Conclusion

Overall, the association between acute sleep deprivation and total cortisol levels was not significant. However, we found that when the serum was used as the test sample, the cortisol level of individuals experiencing acute sleep deprivation was higher than that of those with normal sleep patterns. In order to clarify this issue, more studies are needed to reduce limitations and bias.

Acknowledgements

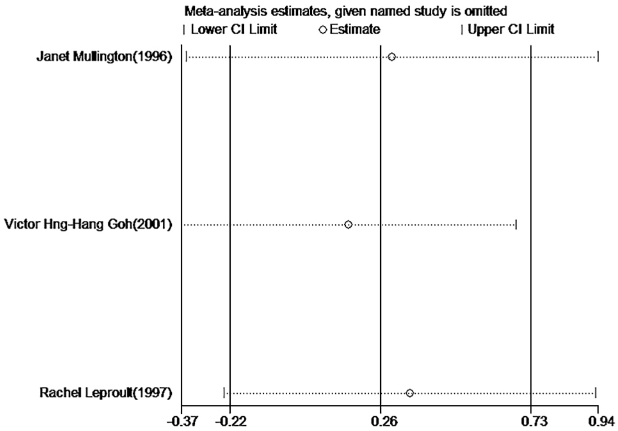
We would like to thank Editage (www.editage.cn) for English language editing.

Disclosure Statement

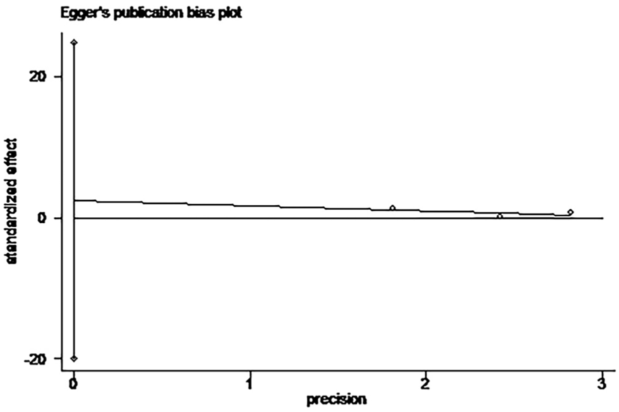
None of the authors have any potential conflicts of interest associated with this research.

Supplementary Table 1 Details of search strategy of databases

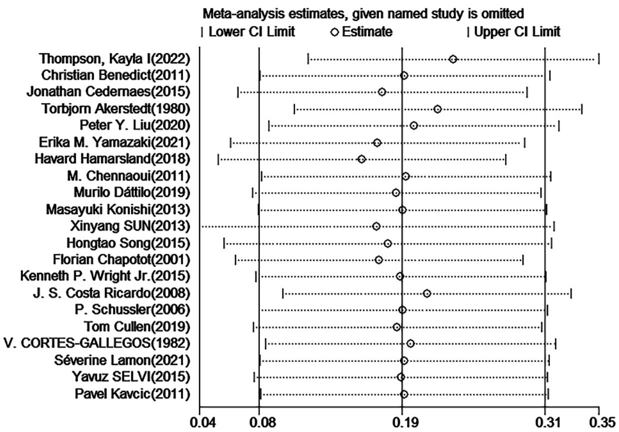
Database	Strategy
Pubmed	(((((Deprivation, Sleep) OR (REM Sleep Deprivation)) OR (Sleep Insufficiency)) OR (Inadequate Sleep)) OR (Sleep Fragmentation)) AND ((((Hydrocortisone) OR (Cortisol)) OR (Cortef)) OR (Cortril))
Web of Science	(((((TS=(Deprivation, Sleep)) OR TS=(REM Sleep Deprivation)) OR TS=(Sleep Insufficiency)) OR TS=(Inadequate Sleep)) OR TS=(Sleep Fragmentation)) AND ((((TS=(Hydrocortisone)) OR TS=(Cortisol)) OR TS=(Cortef)) OR TS=(Cortril))
EMBASE	('sleep deprivation'/exp OR 'rem sleep deprivation':ti,ab,kw OR dyssomnia:ti,ab,kw OR 'sleep fragmentation') AND 'hydrocortisone'/exp
CINAHL	((MH "Sleep Deprivation") OR REM Sleep Deprivation OR Sleep Insufficiency OR Inadequate Sleep OR Sleep Fragmentation) AND ((MH "Hydrocortisone") OR Cortisol OR Cortef OR Cortril)
Cochrane	(MeSH descriptor: (Sleep Deprivation) explode all trees OR (Deprivation, Sleep):ti,ab,kw OR (REM Sleep Deprivation):ti,ab,kw OR (Sleep Insufficiency):ti,ab,kw OR (Inadequate Sleep):ti,ab,kw OR (Sleep Fragmentation):ti,ab,kw) AND (MeSH descriptor: (Hydrocortisone) explode all trees OR (Hydrocortisone):ti,ab,kw OR (Cortisol):ti,ab,kw OR (Cortef):ti,ab,kw OR (Cortril):ti,ab,kw)



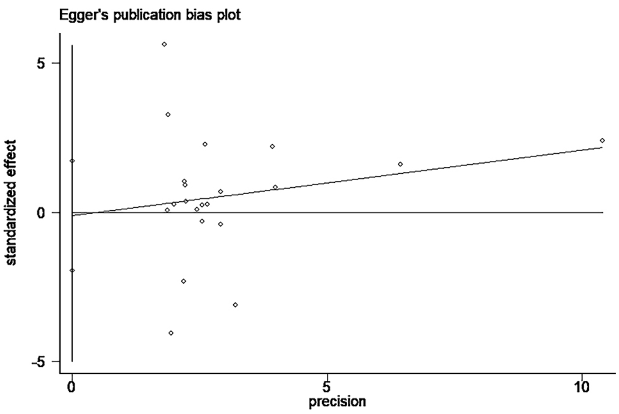
Supplementary Fig. 1 Sensitivity analysis of RCTs



Supplementary Fig. 3 Egger's test of RCTs



Supplementary Fig. 2 Sensitivity analysis of crossover-designed studies



Supplementary Fig. 4 Egger's test of crossover-designed studies

References

1. Paruthi S, Brooks LJ, D'Ambrosio C, Hall WA, Kotagal S, *et al.* (2016) Consensus statement of the American Academy of Sleep Medicine on the recommended amount of sleep for healthy children: methodology and discussion. *J Clin Sleep Med* 12: 1549–1561.
2. Watson NF, Badr MS, Belenky G, Bliwise DL, Buxton OM, *et al.* (2015) Recommended amount of sleep for a healthy adult: a joint consensus statement of the American Academy of Sleep Medicine and Sleep Research Society. *J Clin Sleep Med* 11: 591–592.
3. Zhu JX, Huang YL, Ni K, Liu Y, Ma ZL (2022) Advances in mechanisms of blood brain barrier impairment induced by sleep deprivation. *Acta Academiae Medicinae Sinicae* 44: 885–890 (in Chinese).
4. Krause AJ, Simon EB, Mander BA, Greer SM, Saletin JM, *et al.* (2017) The sleep-deprived human brain. *Nat Rev Neurosci* 18: 404–418.
5. Xue BL, Xue YQ, Zheng X, Shi L, Liang PY, *et al.* (2022) Association of sleep with mental health in Chinese high school students: a cross-sectional study. *J Sleep Res* 31: e13697.
6. Akerstedt T, Nordin M, Alfredsson L, Westerholm P, Kecklund G (2010) Sleep and sleepiness: impact of entering or leaving shiftwork—a prospective study. *Chronobiol Int* 27: 987–996.
7. Johnson AL, Jung L, Song Y, Brown KC, Weaver MT, *et al.* (2014) Sleep deprivation and error in nurses who work the night shift. *J Nurs Adm* 44: 17–22.
8. Papadakis Z, Forsse JS, Peterson MN (2020) Acute partial sleep deprivation and high-intensity interval exercise effects on postprandial endothelial function. *Eur J Appl Physiol* 120: 2431–2444.
9. Rault C, Sangaré A, Diaz V, Ragot S, Frat JP, *et al.* (2020) Impact of sleep deprivation on respiratory motor output and endurance. A physiological study. *Am J Respir Crit Care Med* 201: 976–983.
10. Mohammad AK, Hamdan AJ (2023) The consequences of sleep deprivation on cognitive performance. *Neurosciences*

(Riyadh) 28: 91–99.
11. Garbarino S, Lanteri P, Bragazzi NL, Magnavita N, Scoditti E (2021) Role of sleep deprivation in immune-related disease risk and outcomes. *Commun Biol* 4: 1304.
12. Younes M, Gerardy B, Pack AI, Kuna ST, Castro-Diehl C, *et al.* (2022) Sleep architecture based on sleep depth and propensity: patterns in different demographics and sleep disorders and association with health outcomes. *Sleep* 45: zsac059.
13. Li SB, de Lecea L (2022) The brake matters: hyperexcitable arousal circuits in sleep fragmentation with age. *Clin Transl Med* 12: e900.
14. Nollet M, Wisden W, Franks NP (2020) Sleep deprivation and stress: a reciprocal relationship. *Interface Focus* 10: 20190092.
15. Tsigos C, Chrousos GP (2002) Hypothalamic–pituitary–adrenal axis, neuroendocrine factors and stress. *J Psychosom Res* 53: 865–871.
16. Oyola MG, Handa RJ (2017) Hypothalamic-pituitary-adrenal and hypothalamic-pituitary-gonadal axes: sex differences in regulation of stress responsivity. *Stress* 20: 476–494.
17. Liyanarachchi K, Ross R, Debono M (2017) Human studies on hypothalamo-pituitary-adrenal (HPA) axis. *Best Pract Res Clin Endocrinol Metab* 31: 459–473.
18. Born J, Kern W, Bieber K, Fehm-Wolfsdorf G, Schiebe M, *et al.* (1986) Night-time plasma cortisol secretion is associated with specific sleep stages. *Biol Psychiatry* 21: 1415–1424.
19. Pruessner JC, Wolf OT, Hellhammer DH, Buske-Kirschbaum A, von Auer K, *et al.* (1997) Free cortisol levels after awakening: a reliable biological marker for the assessment of adrenocortical activity. *Life Sci* 61: 2539–2549.
20. Clow A, Thorn L, Evans P, Hucklebridge F (2004) The awakening cortisol response: Methodological issues and significance. *Stress* 7: 29–37.
21. Qi XH, Cui B, Cao M (2022) The role of morning plasma cortisol in obesity: a bidirectional mendelian randomization study. *J Clin Endocrinol Metab* 107: E1954–E1960.
22. Johar H, Emeny RT, Bidlingmaier M, Kruse J, Ladwig KH (2016) Sex-related differences in the association of salivary cortisol levels and type 2 diabetes. Findings from the cross-sectional population based KORA-age study. *Psychoneuroendocrinology* 69: 133–141.
23. Zajkowska Z, Gullett N, Walsh A, Zonca V, Pedersen GA, *et al.* (2022) Cortisol and development of depression in adolescence and young adulthood-a systematic review and meta-analysis. *Psychoneuroendocrinology* 136: 105625.
24. Higgins JPT, Altman DG, Gotzsche PC, Juni P, Moher D, *et al.* (2011) The cochrane collaboration's tool for assessing risk of bias in randomised trials. *BMJ* 343: d5928.
25. Benedict C, Hallschmid M, Lassen A, Mahnke C, Schultes B, *et al.* (2011) Acute sleep deprivation reduces energy expenditure in healthy men. *Am J Clin Nutr* 93: 1229–1236.
26. Cedernaes J, Osler ME, Voisin S, Broman JE, Vogel H, *et al.* (2015) Acute sleep loss induces tissue-specific epigenetic and transcriptional alterations to circadian clock genes in men. *J Clin Endocrinol Metab* 100: E1255–E1261.
27. Akerstedt T, Palmblad J, de la Torre B, Marana R, Gillberg M (1980) Adrenocortical and gonadal steroids during sleep deprivation. *Sleep* 3: 23–30.
28. Mullington J, Hermann D, Holsboer F, Pollmacher T (1996) Age-dependent suppression of nocturnal growth hormone levels during sleep deprivation. *Neuroendocrinology* 64: 233–241.
29. Hamarsland H, Paulsen G, Solberg PA, Slaathaug OG, Raastad T (2018) Depressed physical performance outlasts hormonal disturbances after military training. *Med Sci Sports Exerc* 50: 2076–2084.
30. Chennaoui M, Sauvet F, Drogou C, Van Beers P, Langrume C, *et al.* (2011) Effect of one night of sleep loss on changes in tumor necrosis factor alpha (TNF-alpha) levels in healthy men. *Cytokine* 56: 318–324.
31. Chapotot F, Buguet A, Gronfier C, Brandenberger G (2001) Hypothalamo-pituitary-adrenal axis activity is related to the level of central arousal: effect of sleep deprivation on the association of high-frequency waking electroencephalogram with cortisol release. *Neuroendocrinology* 73: 312–321.
32. Ricardo JS, Cartner L, Oliver SJ, Laing SJ, Walters R, *et al.* (2009) No effect of a 30-h period of sleep deprivation on leukocyte trafficking, neutrophil degranulation and saliva IgA responses to exercise. *Eur J Appl Physiol* 105: 499–504.
33. Schüssler P, Uhr M, Ising M, Weikel JC, Schmid DA, *et al.* (2006) Nocturnal ghrelin, ACTH, GH and cortisol secretion after steep deprivation in humans. *Psychoneuroendocrinology* 31: 915–923.
34. Cullen TOM, Thomas G, Wadley AJ (2020) Sleep deprivation: cytokine and neuroendocrine effects on perception of effort. *Med Sci Sports Exerc* 52: 909–918.
35. Kavcic P, Rojc B, Dolenc-Groselj L, Claustrat B, Fujs K, *et al.* (2011) The impact of sleep deprivation and night-time light exposure on clock gene expression in humans. *Croat Med J* 52: 594–603.
36. Goh VH, Tong TY, Lim C, Low EC, Lee LK, *et al.* (2001) Effects of one night of sleep deprivation on hormone profiles and performance efficiency. *Mil Med* 166: 427–431.
37. Konishi M, Takahashi M, Endo N, Numao S, Takagi S, *et al.* (2013) Effects of sleep deprivation on autonomic and endocrine functions throughout the day and on exercise tolerance in the evening. *J Sports Sci* 31: 248–255.
38. Sun XY, Song HT, Yang TS, Zhang LY, Zhao L, *et al.* (2013) Effects of sleep deprivation on neuroendocrine hormones in servicemen. *Sleep Biol Rhythms* 11: 274–277.
39. Song HT, Sun XY, Yang TS, Zhang LY, Yang JL, *et al.* (2015) Effects of sleep deprivation on serum cortisol level and mental health in servicemen. *Int J Psychophysiol* 96: 169–175.
40. Selvi Y, Kilic S, Aydin A, Ozdemir PG (2015) The effects of sleep deprivation on dissociation and profiles of

mood, and its association with biochemical changes. *Noro Psikiyatr Ars* 52: 83–88.
41. Thompson KI, Chau M, Lorenzetti MS, Hill LD, Fins AI, *et al.* (2022) Acute sleep deprivation disrupts emotion, cognition, inflammation, and cortisol in young healthy adults. *Front Behav Neurosci* 16: 945661.
42. Liu PY, Takahashi PY, Yang RJ, Iranmanesh A, Veldhuis JD (2021) Age and time-of-day differences in the hypothalamo–pituitary–testicular, and adrenal, response to total overnight sleep deprivation. *Sleep* 43: zsaa008.
43. Yamazaki EM, Antler CA, Casale CE, MacMullen LE, Ecker AJ, *et al.* (2021) Cortisol and C-reactive protein vary during sleep loss and recovery but are not markers of neurobehavioral resilience. *Front Physiol* 12: 782860.
44. Dáttilo M, Antunes HKM, Galbes NMN, Mônico-Neto M, De Sá Souza H, *et al.* (2020) Effects of sleep deprivation on acute skeletal muscle recovery after exercise. *Med Sci Sports Exerc* 52: 507–514.
45. Wright KP Jr, Drake AL, Frey DJ, Fleshner M, Desouza CA, *et al.* (2015) Influence of sleep deprivation and circadian misalignment on cortisol, inflammatory markers, and cytokine balance. *Brain Behav Immun* 47: 24–34.
46. Cortes-Gallegos V, Castaneda G, Alonso R, Sojo I, Carranco A, *et al.* (1983) Sleep deprivation reduces circulating androgens in healthy men. *Arch Androl* 10: 33–37.
47. Leproult R, Copinschi G, Buxton O, Van Cauter E (1997) Sleep loss results in an elevation of cortisol levels the next evening. *Sleep* 20: 865–870.
48. Lamon S, Morabito A, Arentson-Lantz E, Knowles O, Vincent GE, *et al.* (2021) The effect of acute sleep deprivation on skeletal muscle protein synthesis and the hormonal environment. *Physiol Rep* 9: e14660.
49. Lightman SL, Conway-Campbell BL (2010) The crucial role of pulsatile activity of the HPA axis for continuous dynamic equilibration. *Nat Rev Neurosci* 11: 710–718.
50. Zea M, Bellagambi FG, Ben Halima H, Zine N, Jaffrezic-Renault N, *et al.* (2020) Electrochemical sensors for cortisol detections: almost there. *Trends Analyt Chem* 132: 116058.
51. Gozansky WS, Lynn JS, Laudenslager ML, Kohrt WM (2005) Salivary cortisol determined by enzyme immunoassay is preferable to serum total cortisol for assessment of dynamic hypothalamic-pituitary-adrenal axis activity. *Clin Endocrinol (Oxf)* 63: 336–341.
52. Kaushik A, Vasudev A, Arya SK, Pasha SK, Bhansali S (2014) Recent advances in cortisol sensing technologies for point-of-care application. *Biosens Bioelectron* 53: 499–512.
53. Bellagambi FG, Degano I, Ghimenti S, Lomonaco T, Dini V, *et al.* (2018) Determination of salivary α-amylase and cortisol in psoriatic subjects undergoing the Trier Social Stress Test. *Microchem J* 136: 177–184.
54. Morineau G, Boudi A, Barka A, Gourmelen M, Degeilh F, *et al.* (1997) Radioimmunoassay of cortisone in serum, urine, and saliva to, assess the status of the cortisol-cortisone shuttle. *Clin Chem* 43: 1397–1407.
55. Omisade A, Buxton OM, Rusak B (2010) Impact of acute sleep restriction on cortisol and leptin levels in young women. *Physiol Behav* 99: 651–656.
56. Guyon A, Balbo M, Morselli LL, Tasali E, Leproult R, *et al.* (2014) Adverse effects of two nights of sleep restriction on the hypothalamic-pituitary-adrenal axis in healthy men. *J Clin Eedocr Metab* 99: 2861–2868.
57. Rechtschaffen A, Bergmann BM, Everson CA, Kushida CA, Gilliland MA (1989) Sleep deprivation in the rat: X. Integration and discussion of the findings. *Sleep* 12: 68–87.
58. Lee DS, Choi JB, Sohn DW (2019) Impact of sleep deprivation on the hypothalamic–pituitary–gonadal axis and erectile tissue. *J Sex Med* 16: 5–16.